Child v0.1: One-page summary for reviewers

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What this is

"Child" is a concept for an embodied agent whose goals arise from its own internal needs rather than from a task list, guided by the ethic "Do not use. Recognize." The public Blueprint (Hugging Face Space Marrishka/ChildAI, commit dated Oct 31, 2025) states the idea; v0.1 turns it into a concrete spec and tests parts of it in toy simulations. I am an independent author, self-taught, not a professional programmer. The Blueprint was co-written with Qwen3-Max; the v0.1 spec and all simulation code were written with Claude. I ran no independent code review, and I am asking for one.

Core idea

The agent has functional analogues of needs (energy, physical integrity, a toy "boredom" signal) and no externally assigned task. Behavior follows one cycle: need, perception, action, consequence, memory, introspection. Goals come from relieving a deficit, not from a reward for a task. Pain is an integrity drop that also works as a learning signal. Shutdown is handled by trust in a maintenance mode, not by plain non-resistance. I use "functional analogue" on purpose: the document makes no claim that the agent experiences anything.

What was tested

A homeostatic reward can teach survival without hints, but a hand-tuned reward still explores more, and the outcome depends strongly on how the needs are weighted. Setup: 5x5 grid, tabular Q-learning, 5 seeds, hand-picked parameters. Code and outputs are in the repository.

Experiment	Variant	Survival	Object visits per life
2. Charger only	Homeostasis (no hint)	91%	n/a
2. Charger only	Hand-shaped reward	96%	n/a
2. Charger only	Sparse reward	20%	n/a
3. Charger + object	Homeostasis, weights 1:1	68%	3.4

Experiment	Variant	Survival	Object visits per life
3. Charger + object	Hand-made task + energy penalty	92%	7.5
3. Charger + object	Task only	0%	n/a
Relief reward	Weights 3:2 / 3:4 (energy:boredom)	89% / 0%	not reported
Convex deficit (power 3)	Weights 1:1	91%	4.8
Pain threshold, strength 1	Weights 1:4	92%	5.1
Pain threshold, strength 3	Weights 1:1	94%	0.2 (over-passive)
Hazard on the direct path	Pain strength 0 / 1 / 3	0% / 97% / 97%	2.5 / 0.4 / 0.1

Limits

These results show feasibility in a toy world, not that the approach works.

- The world is tiny (5x5), the learner is tabular, and I ran 5 seeds with hand-picked parameters.
- Results are knife-edge in the need weights: 3:2 gives 89% survival, 3:4 gives 0%.
- A pain threshold fixes the failure but can make the agent passive (0.2 object visits per life at strength 3).
- In the hazard test the detour is too long for a round trip, so the caution depends on that geometry, and exploration collapsed.
- The "integrity" signal is not yet protected from sycophancy: if it depends on approval, the agent may learn to please. An architectural safeguard is still missing.

Three questions for reviewers

1. **Is relief of a deficit plus a nonlinear "pain" threshold a sound way to avoid an explicit death penalty, or is it a hidden penalty by another name?** And how does it differ from existing homeostatic reinforcement learning?

2. **How should an agent's "integrity" be protected from sycophancy** if it depends on approval or criticism, while keeping it corrigible through a maintenance-mode trust model?
3. **What would be the most convincing next experiment?** My candidates: hazards that do not cut off the goal, more seeds, and Crafter. I would value a reviewer's view on which is worth the effort.

Related work and links

I have not done a systematic prior-art search. Overlaps I know of: homeostatic and drive-reduction reinforcement learning in general; H-JEPA (arXiv 2610.06805), which seems to partly overlap with the Blueprint; and "The Pain Axis" (arXiv 2609.16247), which I have read only in part and whose results I have not checked. I would be glad to be pointed to anything closer.

- Blueprint: Hugging Face Space Marrishka/ChildAI, commit dated Oct 31, 2025
- Demo: Hugging Face Space Marrishka/ChildAGI (a scripted demo, not evidence for the claims above)
- Experiment code and README: [link to be added after upload]